

Response to Reviewer 2

We thank the reviewer for reviewing our work and giving thoughtful and constructive comments. We have made several changes that you have suggested, and we detail specifically how we have addressed your comments below.

Comment 1

In the U.S. dairy industry, what is the approximate proportion of commercial dairy breeders who specialize in breeding and sell bull semen as the main business? And what is the proportion of dairy farmers who breed (most of) their cattle and raise them for dairy products? How does the split of the two types of businesses in the authors' data set compared to that for the entire U.S. industry? Do the two types of business managers differ significantly in their decisions on breeding (e.g., for the most profitable animal genetics to sell vs. for most productive animals to raise) and use of genomic testing (e.g., to grab market share vs. to avoid inbreeding)?

Reply: In our revision, we have clarified that dairy breeders are completely distinct from dairy farmers. In the dairy industry, it is very rare to see commercial breeders also engaging in dairy farming, as stud farms typically specialize in breeding only. We have clarified this on pg. 8:

To describe the dynamics of the dairy industry, we present a simplified theoretical model that describes the coordination problem inherent to dairy cow breeding. In the modern dairy industry, breeding and selecting new bulls for sale has become almost exclusively the job of dairy breeders. In contrast, dairy farmers are largely uninvolved with the breeding process and buy their genetics from these companies instead of producing it themselves or selling genetics. Thus, we model the dairy breeder as a single breeding enterprise and not a farm also engaged in milk production.

Comment 2

Prior to the introduction of genomic testing, why were some lines of bulls more popular than others even though they had comparable or same genetic traits (page 15, top line)? Was the popularity based upon prices, actual productivity (as opposed to the measured genetic traits), branding effects, or something else? After the introduction of genomic testing, did any already popular lines (e.g., super-sires) become less popular and vice versa (because buyers found out some already popular lines now have higher inbreeding rates)? Is it possible for breeders to line breed different lines of cattle since 2008 as some lines ascended to super-sires while other lines became less popular? In other words, can we guarantee there were no crossovers between the treatment group and the control group since 2008?

Reply: One element in the dairy farmer's decision is pedigree. While a bull could be functionally the same, if it is not well known by the industry then it may be seen as a more

risky choice. The popular bulls have more descendants and can be assessed using those descendants. While not documented in any literature, we also understand that there are brand effects in the industry given how pedigrees can become popular by winning dairy cow competitions like the World Dairy Expo.

Another key aspect of our new conceptual model postulates that their popularity is precisely what makes them more valuable for inbreeding after genomic testing. This is now explained on pg. 11:

Genomic testing allows dairy breeders to determine the genetic traits of dairy bulls far more quickly than before. Before it was introduced, it took five years to determine a dairy bull's genetic traits. Once the bull reached sexual maturity around the age of one year, the bull's semen would be frozen and sent to several farms to produce offspring from the bull. After about 10 months, the offspring would be born and would take roughly another two years to get to production age. After a few years of production, the bull now has production data from its offspring that can be used to calculate its genetic traits, in our model q .

Genomic testing gathers the same information in a far shorter amount of time by taking a DNA sample of the bull and correlating its individual genes to traits. In terms of our model, genomic testing caused an acceleration in the rate of quality gain, that is an increase in γ_I and γ_O . In the case that they both increase the same amount, there would be no change in d^{ME} since the relative benefits of inbreeding have not changed. However, genomic testing disproportionately benefits inbreeding because it makes identifying beneficial genes easier. In particular, it makes identifying which offspring inherited the genes easier and so isolating the beneficial gene is easier than it was before. Since outcrossing is about expanding the number of genes in the firm's genetic families and not isolating a specific gene, outcrossing benefits far less from genomic testing. Moreover, this isolation works best when there are fewer genes to test for in the population, and so genomic testing can more efficiently isolate a population which shares less genes (meaning a population with a lower genetic diversity).

As for the point of lines becoming more popular over time, we find that our control lines die out and do not have more descendants. This is why we cannot extend our analysis past 2019. We note this in the footnote on pg. 26:

The last period in the sample is 2018 instead of 2020 because we cannot make inferences after 2018 due to a reduction in the number of animals in the control sample; the number of comparable descendants is too small for us to consider them a valid comparison group. This drastic reduction is a consequence of lower fertility rates in the control group; since those sires fathered fewer sons, their sons will beget even fewer grandsons, and after a few years, this group collapses.

Comment 3

Section 2 Conceptual framework examined the two key, sequential decisions facing animal breeders: to line breed or cross breed; to have genomic testing or not. It would be more

informative to establish a unified model that explicitly addresses the interdependence between these two decisions with the objective function of market share maximization (say, to breed animals with the highest trait values relative to the market). In this way, the authors can claim to explore the coordination between individual dairy breeders and the entire industry by analyzing the strategic actions of animal breeders.

Reply: Thank you for this comment. Per your suggestion, Section 2 now contains a conceptual model that, while not a formally solved mathematical model, explains the strategic coordination between animal breeders and how they compete on quality. The most important insights from the model are:

- Dairy breeders compete on providing quality q to the market and the buyers of bulls do not differentiate bulls except on their quality q . This leads breeders competing to produce the highest q possible because it allows them to capture a high market share.
- Dairy breeders can either inbreed (I) or outcross (O). Inbreeding depletes their genetic diversity d but gives a higher gain in q . Conversely, outcrossing improves their own diversity d by using another firm's stock of d .
- Under certain conditions, primarily a high stock of genetic diversity, it is a best response to inbreed (choose I) because choosing outcrossing (O) gives up market share.
- The primary market friction is that choosing to outcross improves the firm's own d **and** the value of choosing O for the other firms. The first benefit is internalized by the firm but the second is not, and so a social planner would choose to outcross more than the individual firm would in equilibrium. The fact that the firm chooses to inbreed more than the social planner means there is a welfare loss, as the industry's level of genetic diversity (d^{ME}) will be lower than what the social planner would choose (d^*).
- We characterize the effects of genomic testing as increasing the gains from inbreeding relative to outcrossing. This leads to an equilibrium where genetic diversity is even lower (because of more inbreeding) which may or may not make the welfare loss even larger.

We have also renamed the paper to emphasize the importance of the problem being a coordination failure: "Competing for Quality: Coordination Failure in Dairy Genetics Markets"

Comment 4

Regarding the cost-benefit analysis in section 6, the authors admitted that this is only a rough, back-of-the-envelope calculation, although there are still potentially substantial benefits of genomic testing to be included. Once animal breeders decide to have their animals genomically tested, they can ascertain the animals' traits much sooner without waiting an additional three or four years for the daughter-proven results. Meanwhile, animal breeders need to pay about \$50-100 per animal for testing and possibly cull the undesirable animals at a loss. The adjusted net benefits of genomic testing for the entire industry are likely to be greater than originally estimated, which may render inbreeding a lesser issue in terms of the

economic value.

Reply: We agree with your comment that the primary benefits of genomic testing are the ability to ascertain traits quickly (mentioned on pg. 11) and the ability to cull unproductive animals quicker. In our conceptual model, we represent this as increasing the returns to genetic diversity substantially and, most importantly, increasing the returns to inbreeding relative to outcrossing. We discuss this on pg. 12:

The primary prediction of this model is that the introduction of genomic testing increases inbreeding, which potentially exacerbates a welfare loss in the industry. The welfare loss is derived from the fact that breeders excessively inbreed to stay ahead of their competitors, despite the fact that this depletes genetic diversity and leads to less growth in aggregate quality over time. In this paper, our aim is to test one prediction of the model: that genomic testing increases inbreeding. Specifically, we hypothesize that, after genomic testing, the lines that will become the most inbred are lines that were the most popular before genomic testing. Since most bulls would share DNA with these lines, testing their DNA would gain the most information for firms. Taking our model to data, we would then expect inbreeding to increase the most in the popular lines irrespective of quality, as genomic testing will have made inbreeding more beneficial on these lines than before. This would be consistent with the returns to inbreeding increasing relative to outcrossing.

Since our paper cannot properly do a full cost-benefit analysis of genomic testing, we have rewritten the paper to focus on figuring out how much inbreeding has been increased in popular lines relative to lines where inbreeding would not become more valuable. Our cost-benefit analysis receives a mention in the new Section 6 Discussion but the details of it are now in Appendices A and B.

In those sections, we have clarified that the returns to genomic testing are seen in the increase in bull quality, measured by the index Net Merit. Since Net Merit already is adjusted for inbreeding, this means that the increase in quality is higher than the losses from inbreeding depression. However, we also emphasize that the costs of genomic testing are from a lower level of diversity that, compared to the social optimum, delivers less growth in the long run. We explain this now on pg. 31:

Published values of Net Merit are already adjusted for inbreeding, the quality increase will be higher than the losses from inbreeding depression. A more accurate estimation of the benefits requires additional information than the one we have, we need information on the expected future inbreeding for all animals in our sample. We do not have the information required to perform such a calculation.

Comment 5

I would also suggest adding more justifications when the authors try to project the estimated effects on inbreeding and net merit value using the authors' sample data into the U.S. dairy

industry. Related to my comment #1 above, the adoption rates of genomic testing may vary between the (commercial) cattle breeders included in this study and cattle producers in the country, so the projected net benefits of genomic testing for the industry will have to be revised accordingly.

Reply: As we explain in the reply to the first comment, commercial cattle breeders and dairy producers are almost completely distinct businesses in the dairy industry. Moreover, the adoption rate of genomic testing is very high in the industry, and the aggregate benefits are evident in the increase in bull quality over time.

Comment 6

1. Page 2, end of paragraph 1: “industr” should be “industry”.

Reply: the typo was corrected.

2. Page 7, line 4 from bottom: a period is missing before Melton et al.

Reply: the typo was corrected.

3. Page 12, 2nd paragraph, line 3 from the paragraph end: add some numbers to justify “a substantial portion”.

Reply: The paragraph now mentions that the NAAB represents 95% of all livestock genetic companies in the United States.

4. Page 14, 1st paragraph of section 4, lines 4-5: Why are bulls that are NOT genomically tested more likely to have better quality traits and be more widely used?

Reply: The assertion is wrong; it was removed from the text.

5. Page 16, line 4 from page bottom: Figure 5 in the main text is referred to here. Then drop “in the Appendix”.

Reply: the typo was corrected.

6. Page 16, last sentence of the page: perhaps to conduct formal statistical tests on the similarities between the three distributions: matched, treat, and unmatched.

Reply: A balance table was included, as a response to a previous comment.

7. Page 24, line 3 below Figure 8: “percent points” should be “percentage points”.

Reply: the typo was corrected.